6 A filament lamp is connected in series to a cell of e.m.f. 4.0 V and negligible internal resistance and a variable resistor as shown in Fig. 6.1. The variation with the potential difference V across the filament lamp of the current I is as shown in Fig. 6.2.

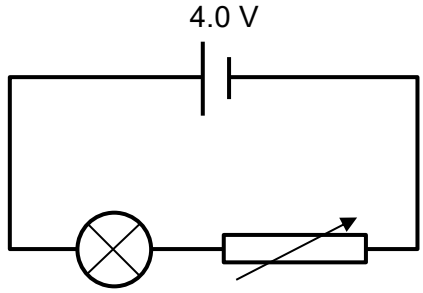

Fig. 6.1

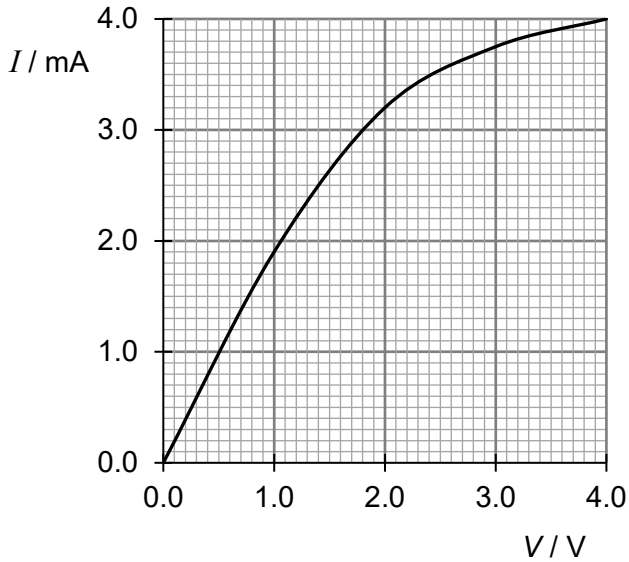

Fig. 6.2

(a) Determine the resistance of the variable resistor when V is 1.4 V.

resistance = ... Ω [2]

(b) On Fig. 6.3, sketch a labelled graph of the resistance R of the filament lamp against V between V = 1.0 V and 4.0 V.

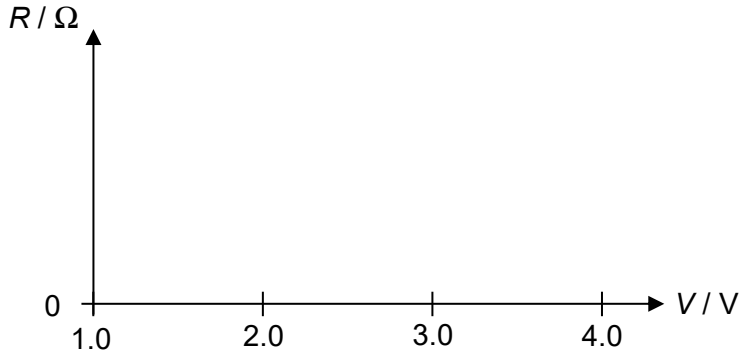

Fig. 6.3

[2]

(c) Discuss how the resistance of a filament lamp changes with temperature.

.....

.....

..... [2]